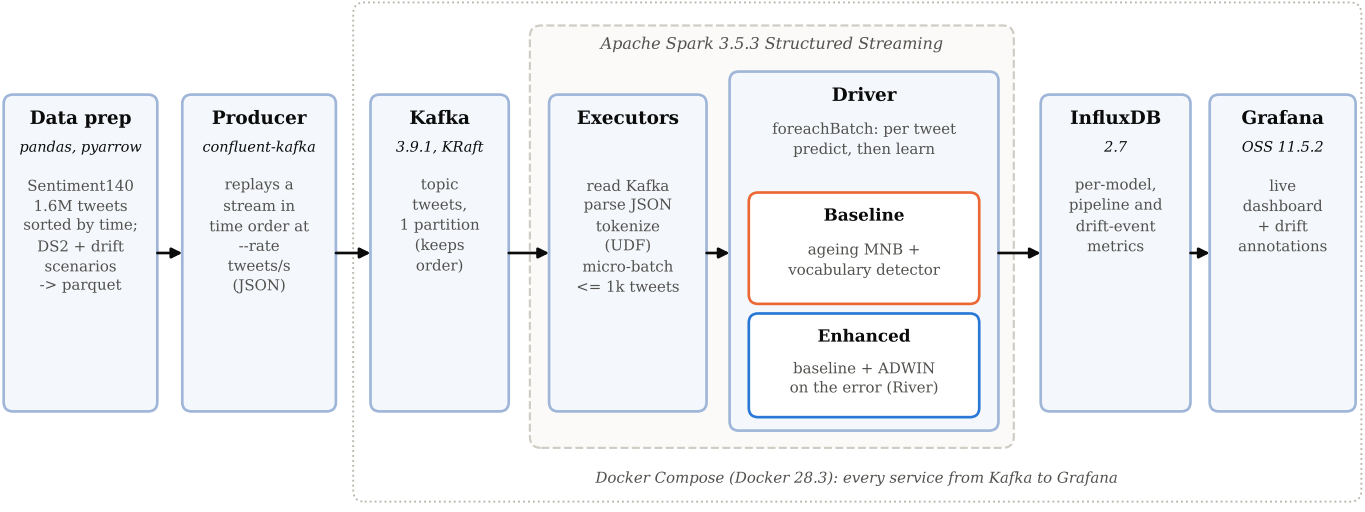
Data prep
pandas, pyarrow
Sentiment140
1.6M tweets
sorted by time;
DS2 + drift
scenarios
-> parquet
Producer
confluent-kafka
replays a
stream in
time order at
--rate
tweets/s
(JSON)
Kafka
3.9.1, KRaft
topic
tweets,
1 partition
(keeps
order)
Apache Spark 3.5.3 Structured Streaming
Executors
read Kafka
parse JSON
tokenize
(UDF)
micro-batch
<= 1k tweets
Driver
foreachBatch: per tweet
predict, then learn
Baseline
ageing MNB +
vocabulary detector
Enhanced
baseline + ADWIN
on the error (River)
InfluxDB
2.7
per-model,
pipeline and
drift-event
metrics
Grafana
OSS 11.5.2
live
dashboard
+ drift
annotations
Docker Compose (Docker 28.3): every service from Kafka to Grafana